

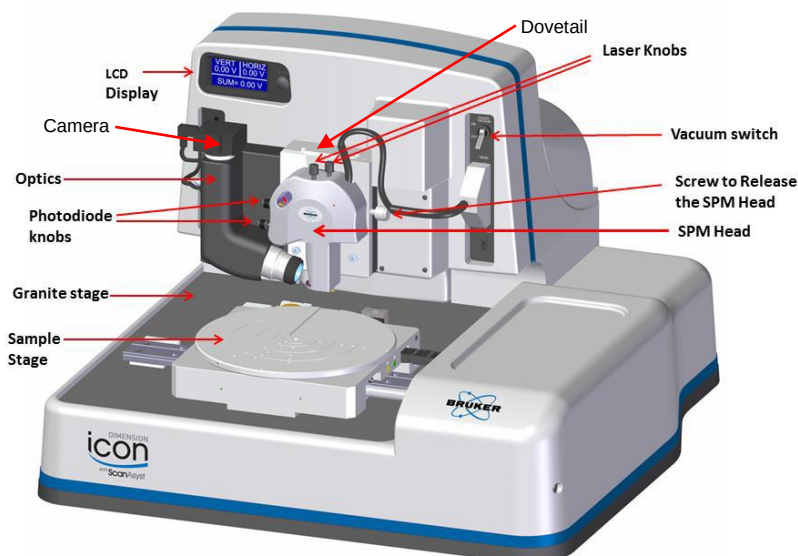
Bruker Dimension Icon SPM SOP

- NanoScope V SPM Control Station
- Icon XYZ Closed-Loop SPM Scan Head with nominal range of $90\text{ }\mu\text{m} \times 90\text{ }\mu\text{m}$ and vertical range of $10\text{ }\mu\text{m}$

Admins:

In case of **tip change** or issues please contact:

- 1) ~~Mdm Ng SN (6516 2823)~~
- 2) ~~Mr Ong PM (6516 2632)~~



I. FACTORS TO TAKE INTO ACCOUNT BEFORE USING THE SPM - **DO's and DON'Ts**

- 1) HANDLE THE SPM SCANNER HEAD WITH CARE. The scanner head costs \$60,000!! (check there is enough clearance between the scanner tip end and the stage/sample surface)
- 2) **DO** raise the SPM head to a clear position before moving the stage in XY
- 3) **DO NOT** hit the scanner end cap with other objects
- 4) **DO NOT** drop the head into the dovetail
- 5) **DO NOT** touch the lens on the SPM head
- 6) **DO NOT** CHANGE THE TIP ON YOUR OWN EVENTHOUGH YOU KNOW HOW TO DO IT.
- 7) **DO NOT** touch the camera
- 8) **DO NOT** stare directly at the laser light
- 9) **DO NOT** PLACE ANYTHING ON THE GRANITE STAGE AT ALL !
- 10) **DO NOT** turn off the PC. Never install any programs or change any settings on the PC.
- 11) **DO** wait while the software is loading processes.
- 12) **DO** set **X and Y offset to ZERO** after withdraw.
- 13) **DO** save your data in T:\ drive
- 14) **DO** copy out your data from Acer PC. Your data may be deleted without warning.
- 15) **DO** put on gloves.

II. INTRODUCTION TO THE BASIC IMAGING MODES OF AFM

A) ScanAsyst

B) Tapping Mode

C) Contact Mode

A) ScanAsyst AFM

ScanAsyst imaging mode is the first automatic image-optimization scanning mode for AFM. ScanAsyst continuously monitors image quality, and makes the appropriate parameter adjustments such as setpoints, feedback gains, and scan rates. ScanAsyst is based on Peak Force Tapping mode which collects force curves at every pixel in the image. The peak force of each of the force curves is used as the imaging feedback signal, hence providing direct force control. This enables it to operate at very low forces than Tapping Mode, which helps protect delicate samples and probes. Peak Force Tapping mode modulates the Dimension Icon Z-piezo at ~2 kHz with a default Peak Force Amplitude of 150 nm (0 to peak). Using ScanAsyst provides faster imaging while retaining high resolution images even for sticky samples.

B) Tapping Mode AFM

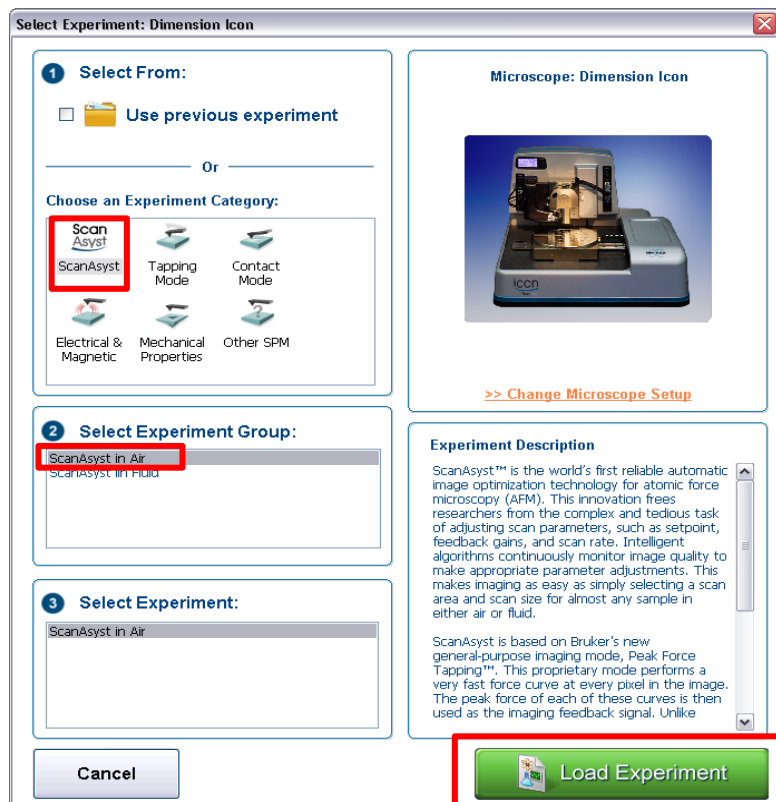
The probe is oscillated near its resonance frequency and it intermittently interacts with the sample surface. Constant oscillation amplitude is used as a feedback mechanism to maintain a constant tip-sample interaction. Tapping mode minimizes shear forces, which can damage delicate samples. Topography and phase images are simultaneously acquired with tapping mode AFM. Samples which are delicate as well as with higher adhesive properties can be characterized using tapping mode AFM.

C) Contact Mode AFM

Measures topography and lateral force image/friction by scanning the probe's tip in a raster pattern on the sample surface while monitoring the cantilever deflection. Constant deflection between the cantilever and the sample is therefore used as feedback mechanism. In contact mode, the tip continuously remains in contact with the sample.

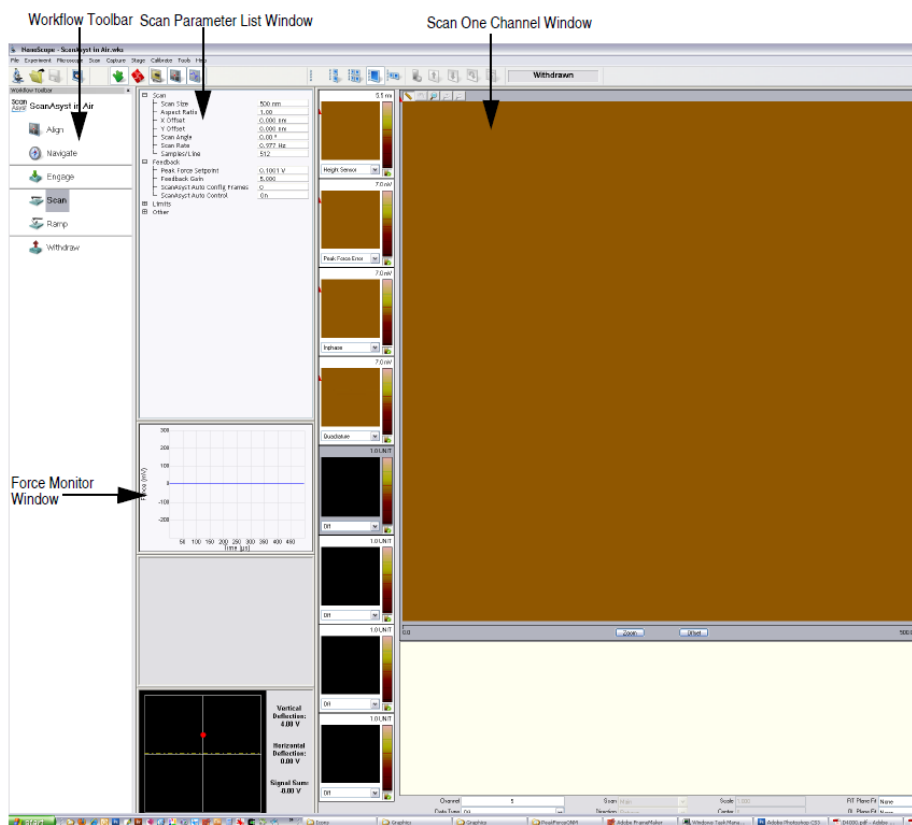
A) ScanAsyst

1. Put on gloves.
2. Turn on N2 gas valve.
3. Move and click the left mouse or turn on power button at bottom right corner of monitor.
4. Double click the **Nanoscope** software  on the desktop.
5. Select **ScanAsyst** in the **Choose an Experiment Category** panel.

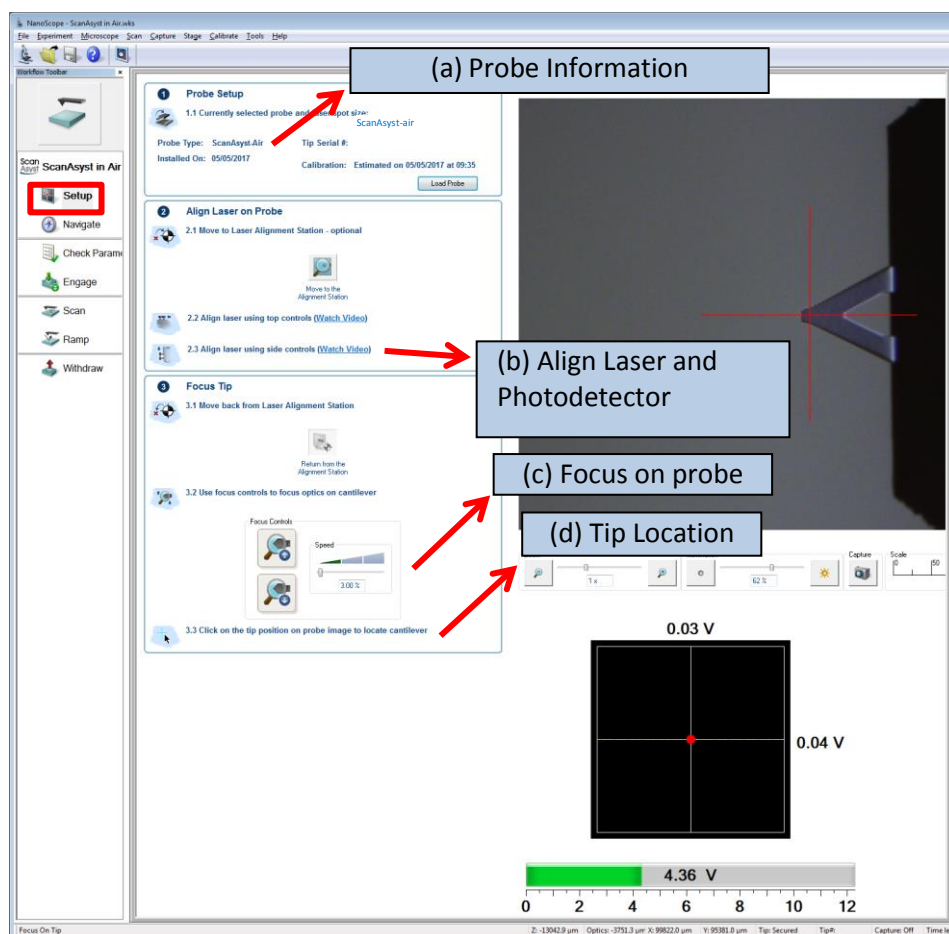


6. Select **ScanAsyst in Air** in the **Select Experiment Group** panel and click **Load Experiment**.

The Workflow Toolbar, Scan Parameters List window, Scan One Channel window and Force Monitor window will open.

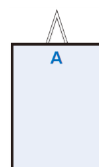
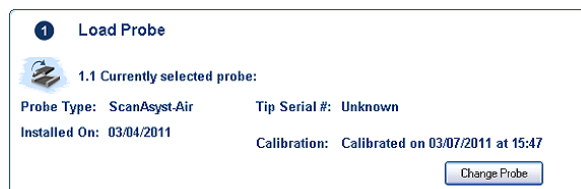


7. Click **Setup** in the Workflow Toolbar.
(Probe information → Align Laser & Adjust photodetector position → Focus Probe → Tip Location)



(a) Probe Information: ScanAsyst-Air (70 kHz, $k \sim 0.4$ N/m, tip radius < 10 nm)

***Always ensure that you use the correct probe (Check with Lab Technologist)**

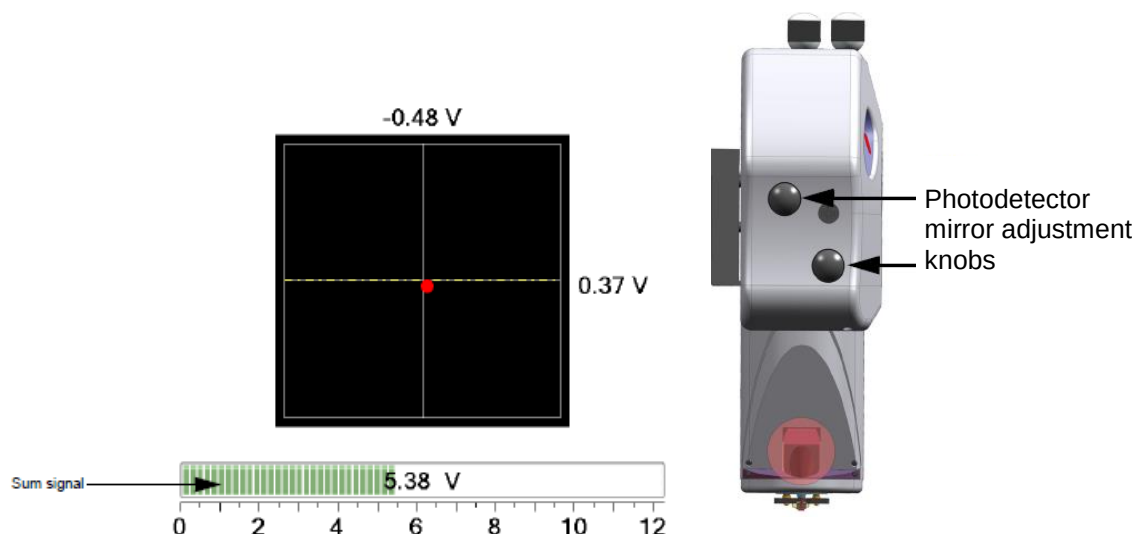
**(b) Align Laser and Photodetector** (Do not turn the knobs too much as mechanism inside the head could break. **NEVER FORCE THE KNOBS.**)

(i) *Laser Alignment.* Skip this step. It should have been pre-aligned. **Scanner head costs ~\$60K!**

Verify that the SUM signal is 4 - 6 V (laser power reaching the detector).

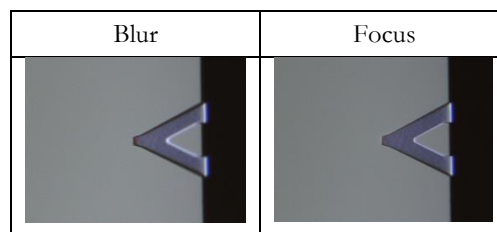
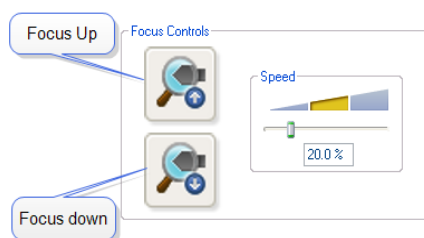
(ii) *Adjust photodetector.*

Verify / use the adjustment knobs located on the left side of the scanner, to center the laser diode on the quad photodetector. Set the Vertical and Horizontal Deflections to 0 V.

**(c) Focus Probe** (**must be done before focusing the optics on sample surface**)

[Optics is focused on probe tip so that Z position of tip is known. This is needed for successful tip engagement on sample surface.]

Adjust the speed accordingly and **Focus Up/down** till the cantilever end is sharp. Adjust illumination for better view.

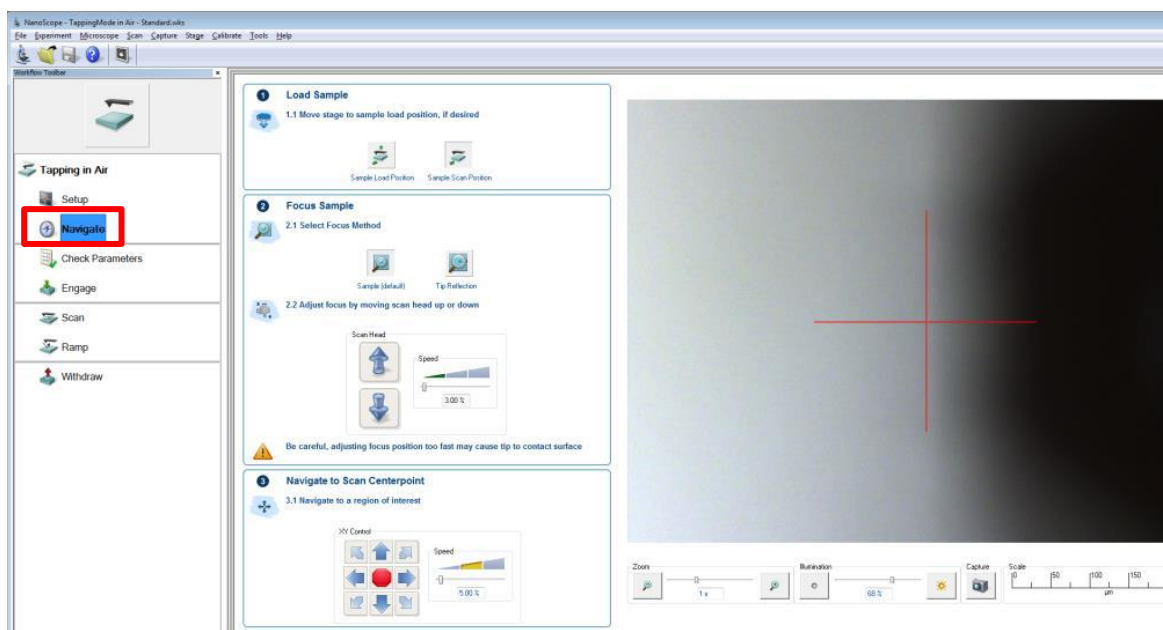
**(d) Tip Location**

Click to mark the crosshair on the position of the tip.

(When you leave the Locate tip mode, the optics move to a focus position, typically 1 mm below the tip.)



8. When the tip is in focus and centered in the field of view, click **Navigate** in the Workflow Toolbar.
(Focus Sample → navigate to area of interest)



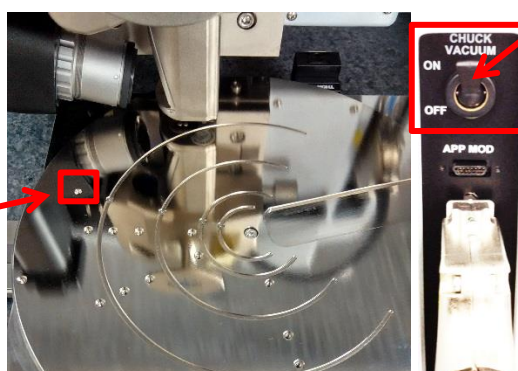
- a) Place sample on the stage and toggle 'ON' the chuck vacuum switch at the upper-right corner of the microscope, so that the sample is held down onto the stage.

*Ensure sample covers the vacuum hole fully and that the bottom of the sample is flat, so that sample is held down using vacuum.

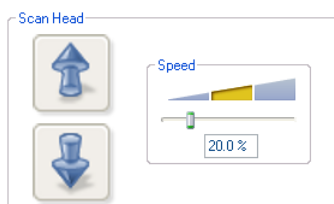
*If the sample size is big, use the other vacuum holes which are meant for big samples.
(Ensure only 1 vacuum hole is open at all times; block all unused holes with screws.)

Toggle switch to turn on vacuum to hold down sample

Vacuum holes
for small sample



- b) **Raise the SPM Head** using Scan Head "up arrow" to make sure that there is enough clearance space between the tip and the sample surface.



CAUTION:

When moving the SPM stage up and down, it is possible to crash the tip into the surface. To prevent a crash while focusing on the surface, watch the optical image and tip-to-sample proximity. The sample should be in focus when the tip is 1mm (1000µm) above the surface.

* For samples which are difficult to bring into focus, move to an edge of the sample, which is easy to find in the optical image, and bring the top of the edge into focus.

- (i) **Carefully** rotate the sample stage, like a turntable, so that the laser light falls on the sample surface.
- (ii) Looking at tip-sample distance, move the tip to 1-2 mm above the sample surface by using the **Scan Head** arrows to focus on the sample surface or by rolling the trackball up or down while pressing the bottom-left button. [When you start seeing the red laser diffraction, on the optical image in the video panel, you are very close to finding the surface. **Decrease** the approach **speed** of the scanner head.]

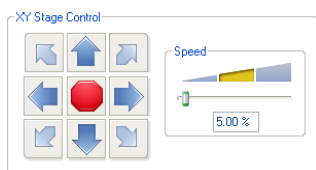


CAUTION:

Because the command **Focus Surface** moves the scanner vertically, be careful when making this adjustment to ensure that the tip does not hit the sample surface.

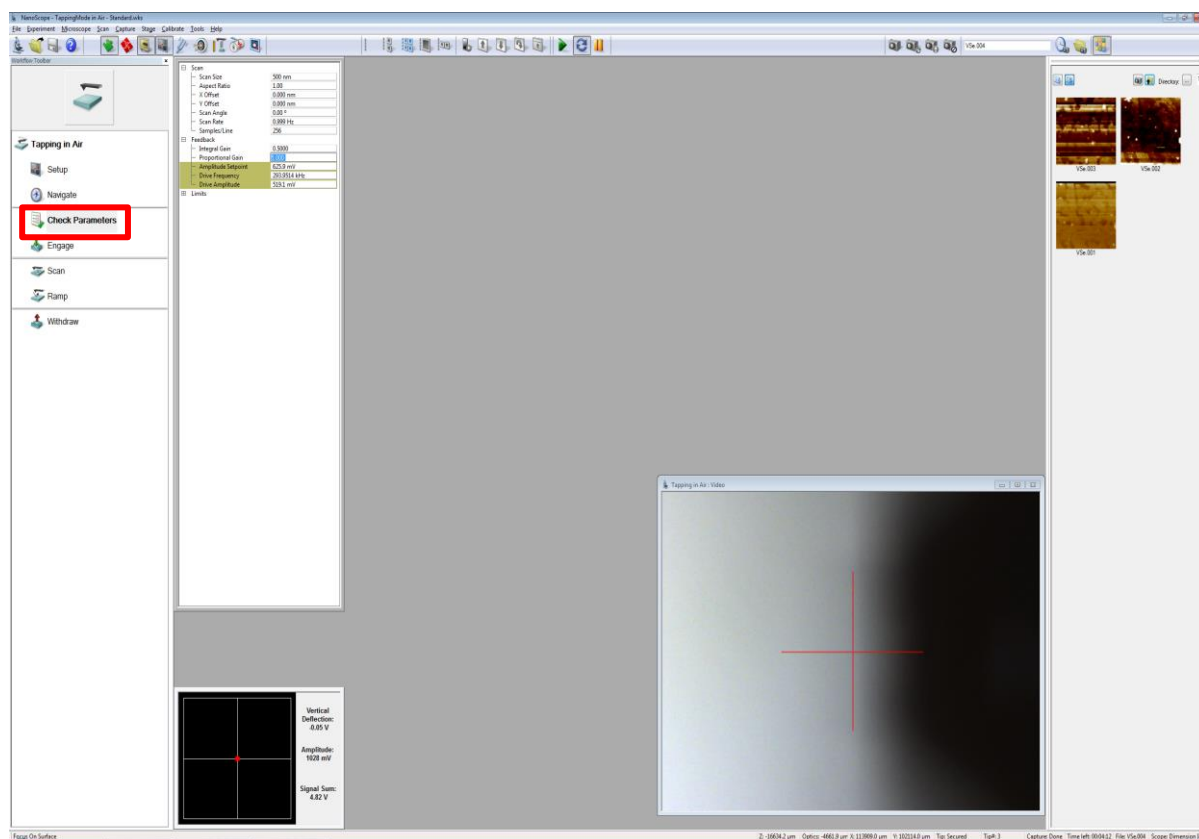
- c) Locate area of interest.

- (i) Move the X-Y stage to align the desired location on the sample under the crosshairs, either by using the XY Stage Control arrows or by rolling the trackball (without holding down any buttons).



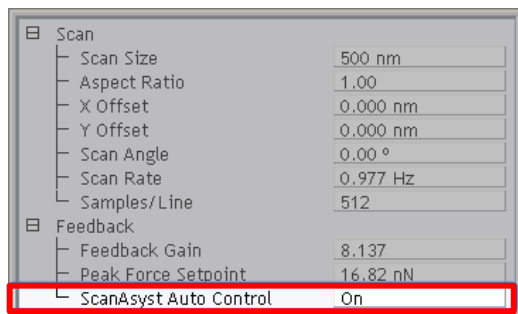
- (ii) Verify that the surface is in focus.

9. Click **Check Parameters** in the Workflow Toolbar.



Bruker recommends always to engage with small scan sizes.

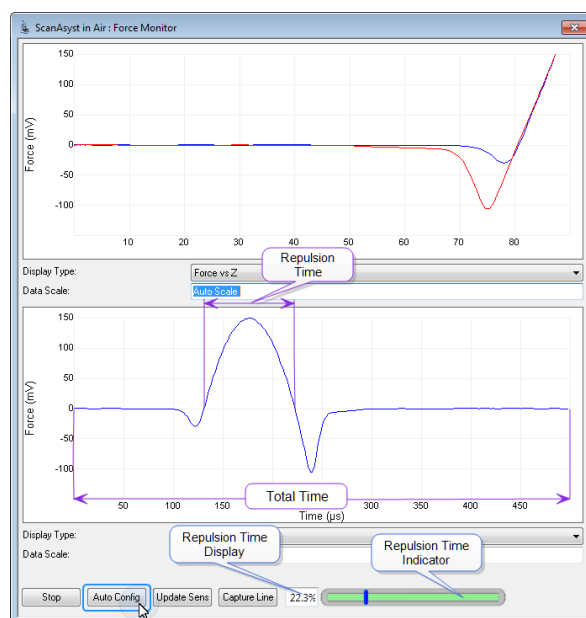
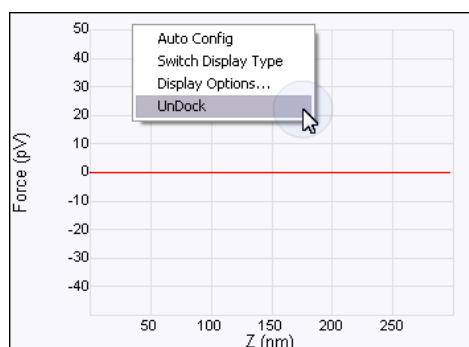
- X and Y Offsets: 0
- Scan Angle: 0
- ScanAsyst Auto Control: **On**



10. Click **Engage** to achieve “smart” approach of the tip.

The software will optimize the image (Scan size changes to 10 nm for about 5 secs). Then the software will adjust the setpoint automatically.

- Set Channels 1 to **Height Sensor** and Channel 2 to **Peak Force Error**.
- Click **AutoScale** icon  or set the Data Scale to a reasonable value.
- Optimizing a ScanAsyst Image
 - UnDock** the force monitor window by right clicking the force monitor panel in the Scan view.

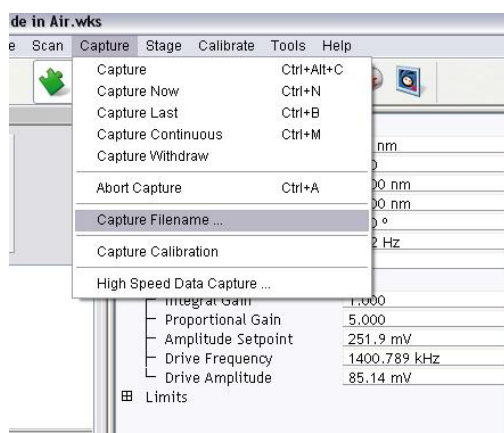


- Click **Auto Config**
A portion of the bar will turn yellow if the Repulsion time is less than 10% of the total time. Manually adjust the Peak Force Setpoint and/or the Peak Force Amplitude to correct this.
- Manually optimizing ScanAsyst Image
Turn **ScanAsyst Auto Control** from **On** to **Individual** to allow manual control.

*Peak Force Setpoint that is too high can either damage the sample or wear the tip.

*Increasing Peak Force Amplitude will reduce the contact time during each tip tapping cycle on the sample and help tracking the rough and/or sticky sample by avoiding a situation where the tip is unable to pull off from the sample.

11. Image Capturing.

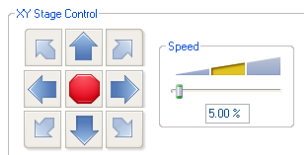


- a) Choose **Capture Filename** from the **Capture** menu and create your folder in T:\ drive.
*Make sure you have set the correct directory.
- b) Type your filename and click **Capture**.
- c) **Capture on** will be shown at the bottom right of the monitor.
*Capture will be completed when the full image is recorded under one set of parameters. If parameters are changed during one recording, the program will capture the next image.

12. To move to another area of interest, execute **Withdraw** in the Workflow Toolbar.

The SPM stops scanning and ascends.

- a) Use Scan Head up arrow (**Navigate** in Workflow Toolbar) if more clearance between the tip and sample is desired.
- b) Move the X-Y stage to align the desired location on the sample under the crosshairs, either by using the XY Stage Control arrows or by rolling the trackball (without holding down any buttons).



- c) Check surface is focused before clicking **Engage** (engage with small scan size, X and Y Offset = 0).

13. SHUTDOWN

- a) Click **Withdraw** in the Workflow Toolbar.
- b) Set X and Y offsets: 0, Scan angle: 0°
- c) Click **Navigate** and raise the scanner head so it is visibly clear of the surface. **LIFT the head up AT LEAST 1 cm above your sample.**



CAUTION:

Never withdraw samples without verifying that the tip has adequate clearance during the entire sample removal sequence.

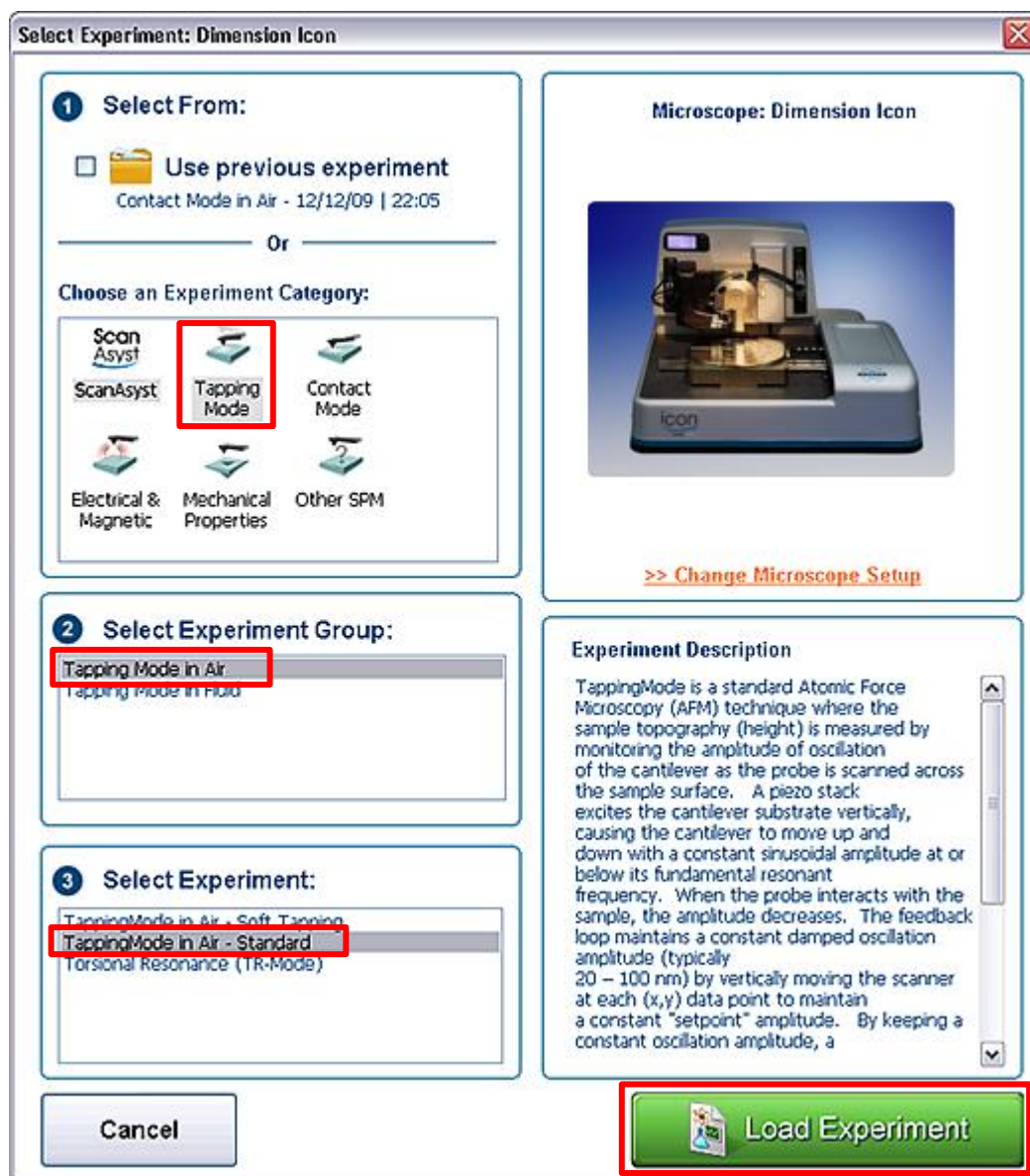
- d) With gloves on, open the window and rotate the stage.
- e) Turn off vacuum and collect sample.



- f) If necessary, clean the stage. Only use isopropanol on a KimWipe and never wet the stage directly with solvent. **DO NOT use acetone.**
- g) Close the glass window.
- h) Close the Nanoscope software.
- i) Turn off the N2 gas valve.
- j) Copy out your data from the Acer PC.
Use NanoScope Analysis Software to Modify image, e.g. Plane Fit, Flatten.
- k) Sign out of the logbook.

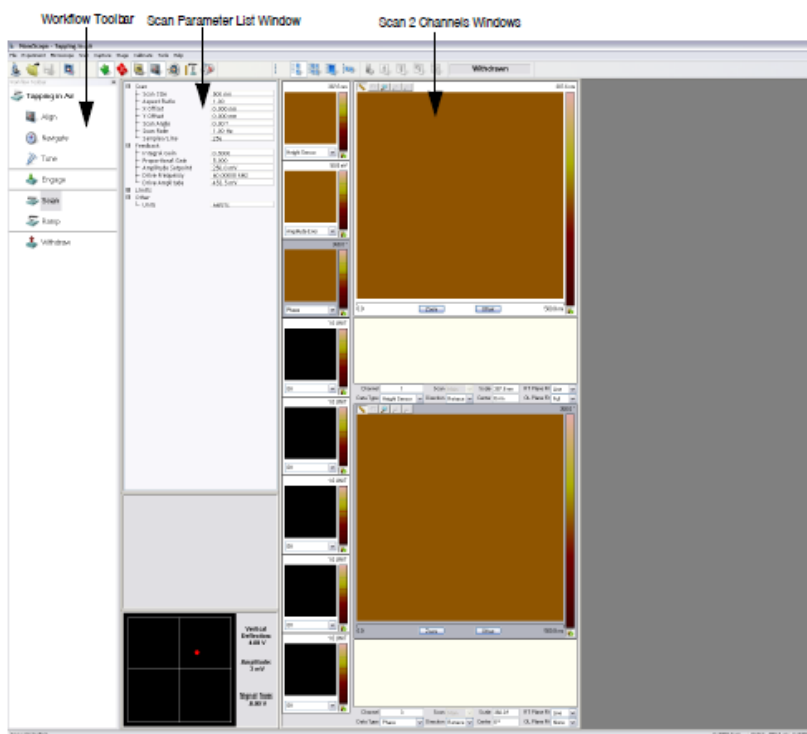
B) Tapping Mode AFM

1. Put on gloves.
2. Turn on N₂ gas valve.
3. Move and click the left mouse or turn on power button at bottom right corner of monitor.
4. Double click the **Nanoscope** software  on the desktop.
5. Select **Tapping Mode** in the **Choose an Experiment Category** panel.

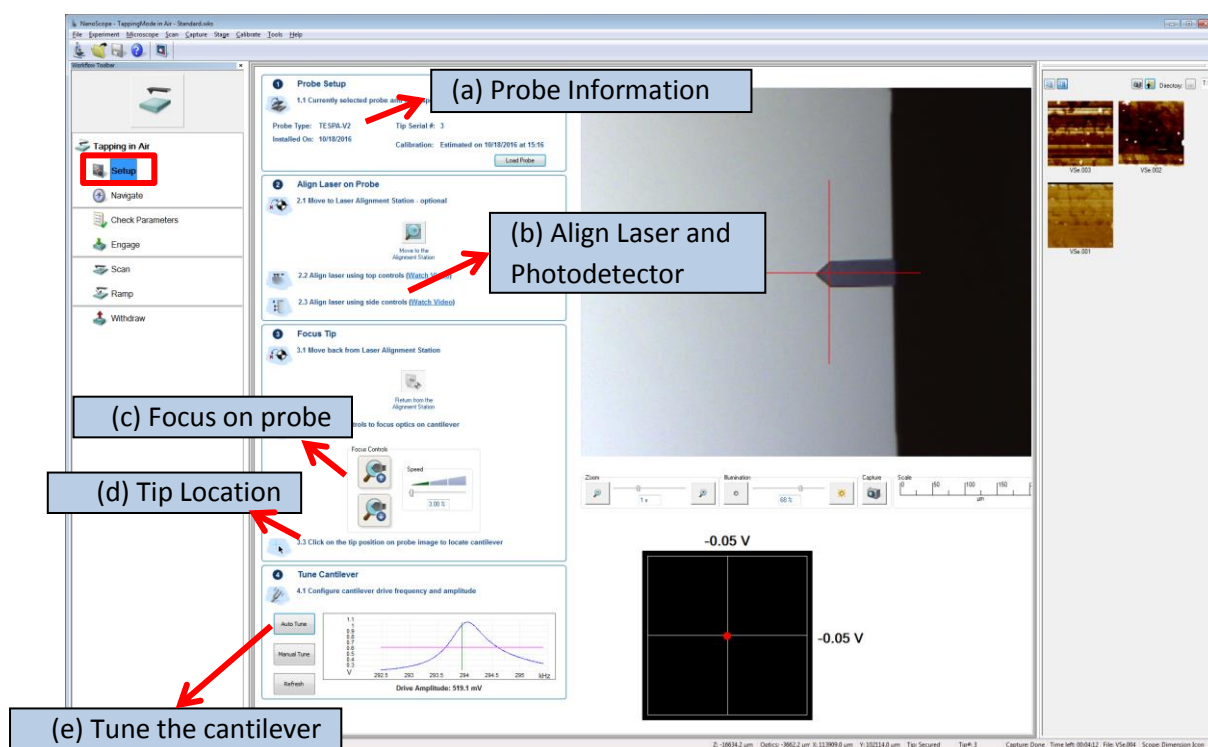


6. Select **Tapping Mode in Air** in the **Select Experiment Group** panel.
7. Select **TappingMode in Air - Standard** in the **Select Experiment** panel and click **Load Experiment**.

The Workflow Toolbar, Scan Parameters List window and Scan 2 Channels windows will open.



8. Click **Setup** in the Workflow Toolbar.
(Probe information>Align Laser & Adjust photodetector position>Focus Probe>Tip Location>Tune Cantilever)



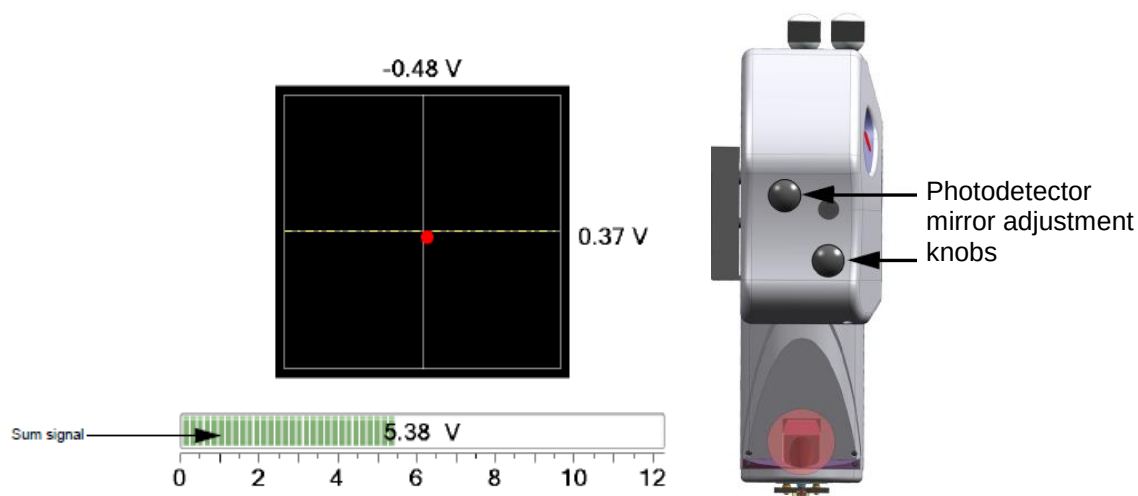
- (a) **Probe Information: TESPA-V2** (320 kHz, $k \sim 37$ N/m, tip radius < 10 nm)

**Always ensure that you use the correct probe (Check with Lab Technologist)*



(b) Align Laser and Photodetector (Do not turn the knobs too much as mechanism inside the head could break. **NEVER FORCE THE KNOBS.**)

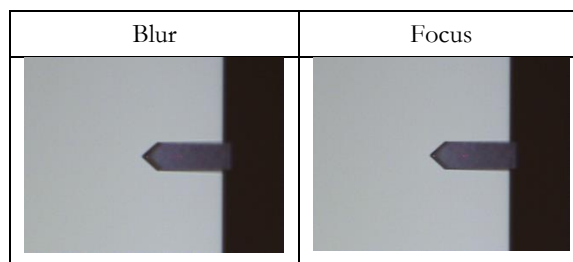
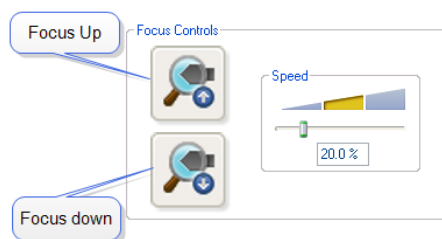
- i) *Laser Alignment.* Skip this step. It should have been pre-aligned. **Scanner head costs ~\$60K!**
Verify that the SUM signal is 4 - 6 V (laser power reaching the detector).
- ii) *Adjust photodetector.*
Verify/ use the adjustment knobs located on the left side of the SPM head, to center the laser diode on the quad photodetector. Set the Vertical and Horizontal Deflections to 0 V.



(c) Focus Probe (**must be done before focusing the optics on sample surface**)

[Optics is focused on probe tip so that Z position of tip is known. This is needed for successful tip engagement on sample surface.]

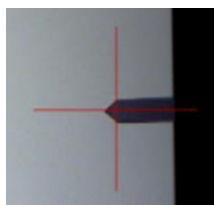
Adjust the speed accordingly and **Focus Up/down** till the cantilever end is sharp. Adjust illumination for better view.



(d) Tip Location

Click to mark the crosshair on the position of the tip.

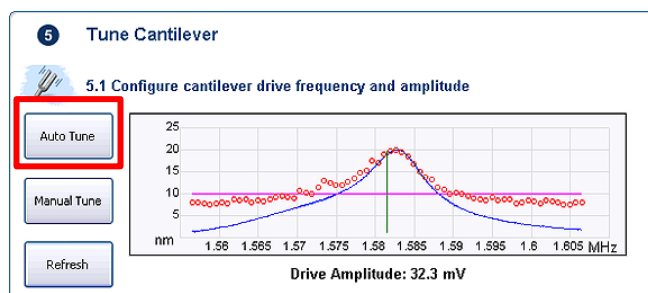
(When you leave the Locate tip mode, the optics move to a focus position, typically 1mm below the tip.)



(e) Tune Cantilever (Only in TappingMode)

(Determine optimal frequency, amplitude and phase of driving signal to drive the cantilever to oscillate at a constant amplitude)

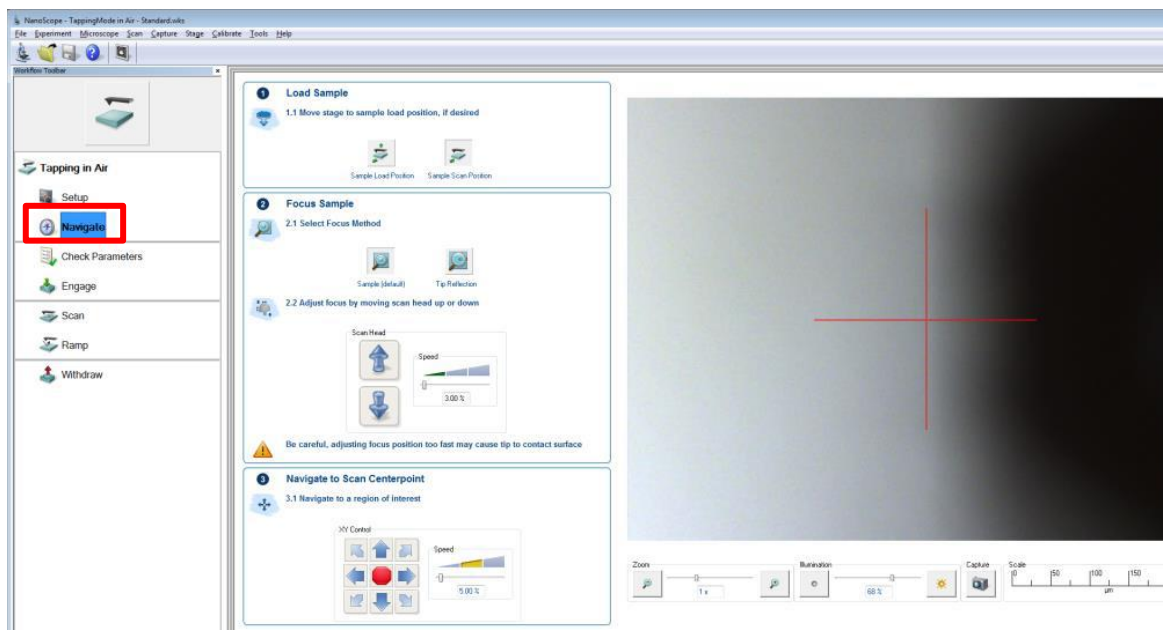
- (i) Click “Auto tune” (For most purposes, the **Auto Tune** option is sufficient)



In most instances, the resonant peak has a sharp Gaussian distribution.

Note: For the system to work well in TappingMode, the drive frequency must be at or below the peak in the resonance plot. The resonant frequency can decrease during engage, due to the short distance between the cantilever and the sample surface. The drive frequency is configured lower than the resonant frequency to avoid the possibility that the resonant frequency might shift below the drive frequency upon approach of the tip to the surface.

9. When the tip is in focus and centered in the field of view, click **Navigate** in the Workflow Toolbar.
(Focus Sample → navigate to area of interest)



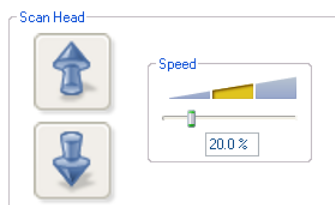
- a) Place sample on the stage and toggle ‘ON’ the chuck vacuum switch at the upper-right corner of the microscope, so that the sample is held down onto the stage.

*Ensure sample covers the vacuum hole fully and that the bottom of the sample is flat, so that sample is held down using vacuum.

*If the sample size is big, use the other vacuum holes which are meant for big samples.
(Ensure only 1 vacuum hole is open at all times; block all unused holes with screws.)



- b) **Raise the SPM Head** using Scan Head “up arrow” to make sure that there is enough clearance space between the tip and the sample surface.

**CAUTION:**

When moving the SPM stage up and down, it is possible to crash the tip into the surface. To prevent a crash while focusing on the surface, watch the optical image and tip-to-sample proximity. The sample should be in focus when the tip is 1mm (1000µm) above the surface.

* For samples which are difficult to bring into focus, move to an edge of the sample, which is easy to find in the optical image, and bring the top of the edge into focus.

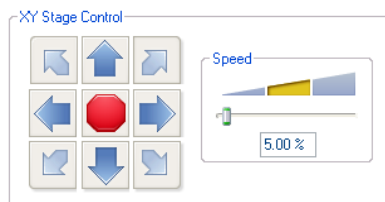
- (i) **Carefully** rotate the sample stage, like a turntable, so that the laser light falls on the sample surface.
- (ii) Looking at tip-sample distance, move the tip to 1-2 mm above the sample surface by using the **Scan Head** arrows to focus on the sample surface or by rolling the trackball up or down while pressing the bottom-left button. [When you start seeing the red laser diffraction, on the optical image in the video panel, you are very close to finding the surface. **Decrease** the approach **speed** of the scanner head.]

**CAUTION:**

Because the command Focus Surface moves the scanner vertically, be careful when making this adjustment to ensure that the tip does not hit the sample surface.

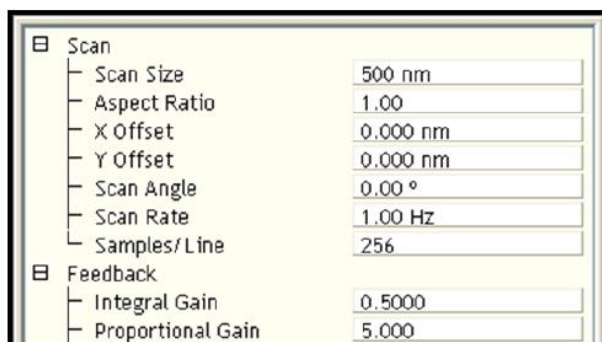
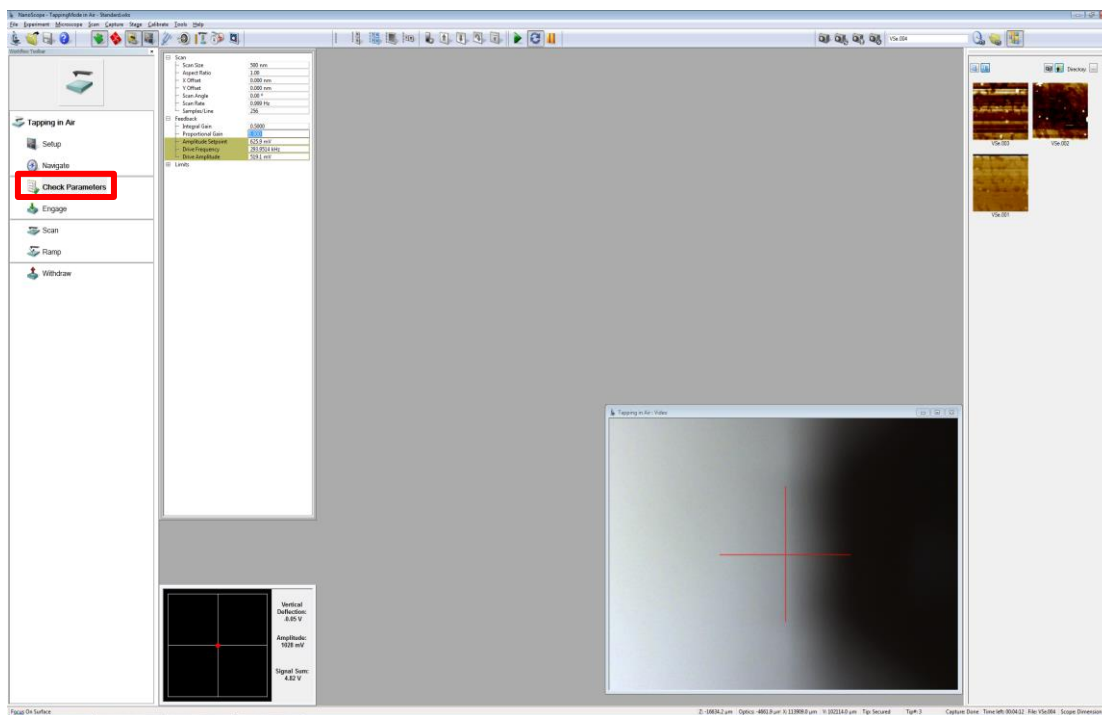
- c) Locate area of interest.

- (i) Move the X-Y stage to align the desired location on the sample under the crosshairs, either by using the XY Stage Control arrows or by rolling the trackball (without holding down any buttons).



- (ii) Verify that the surface is in focus.

10. Click **Check Parameters** in the Workflow Toolbar.



Initial scan parameters for TappingMode in Air

Bruker recommends always to engage with small scan sizes

- Scan Size: 500 nm
- X and Y Offset: 0
- Scan Angle: 0°
- Scan Rate: 1 Hz
- Integral gain: 0.5
- Proportional gain: 5

11. Click **Engage** in the Workflow Toolbar.

Scan lines appear in the Image Windows once the tip engages and scanning begins.

- a) Channel 1 Data Type: **Height Sensor**
Channel 2 Data Type: **Amplitude Error**
Channel 3 Data Type: **Phase**
- b) Click **AutoScale** icon  or set the Data Scale to a reasonable value.
- c) Adjust scan size to desired size.

e) Optimizing Scan Parameters.

Check to see whether the trace and retrace lines in the **Scope** window are tracking each other well. They should have a similar shape, but they may not overlap each other horizontally or vertically. Adjust the **Amplitude Setpoint**, **Integral Gain**, **Proportional Gain**, and **Scan Rate** parameters. Once the trace and retrace are tracking well, the tip is scanning the sample surface.

(i) Adjust **Amplitude Setpoint**.

The force applied by the tip on the sample increases as the setpoint is decreased. To minimize the force on the sample:

Increase the setpoint using the right arrow key on the keyboard, until the tip is no longer interacting with the surface (the trace and retrace lines are different and the Z Center Position move to the retracted limit).

Slowly decrease the amplitude setpoint using the left arrow key until features appear in the trace and retrace lines (the blue and red lines are very close or similar).

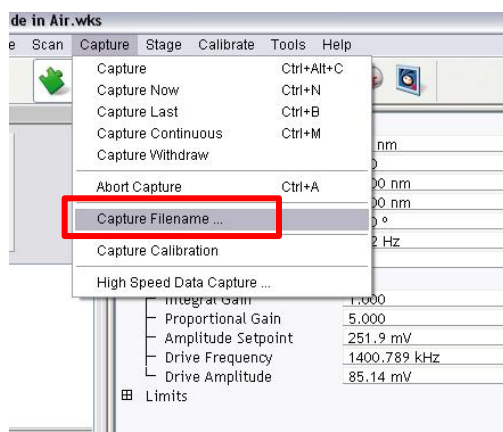
(ii) Adjust **Integral** and **Proportional gains** to minimize error and noise. They must be high enough to force the feedback system to track the sample surface.

Increase **Integral Gain** until the piezo begins to oscillate, then eliminate the oscillations by reducing the gain with 2 - 3 clicks of the left arrow. Repeat for **Proportional Gain** (usually set to 30%-100% more than **Integral Gain**)

(iii) Adjust **Scan Rate**. Decrease the **Scan Rate** as the **Scan Size** is increased.

Use 0.5 - 1.0 Hz for large scans on samples with tall features. High scan rates help reduce drift, but only use them on flat samples with small scan sizes.

12. Image Capturing.

a) Choose **Capture Filename** from the **Capture** menu and create your folder in T:\ drive.

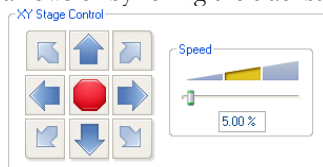
**Make sure you have set the correct directory.*

b) Type your filename and click **Capture**c) **Capture on** will be shown at the bottom right of the monitor.

**Capture will be completed when the full image is recorded under one set of parameters. If parameters are changed during one recording, the program will capture the next image.*

13. To move to another area of interest, execute **Withdraw** in the Workflow Toolbar.

The SPM stops scanning and ascends.

a) Use **Scan Head up** arrow (**Navigate** in Workflow Toolbar) if more clearance between the tip and sample is desired.b) Move the X-Y stage to align the desired location on the sample under the crosshairs, either by using the XY Stage Control arrows or by rolling the trackball (without holding down any buttons).c) Check surface is focused before clicking **Engage** (engage with small scan size, X and Y Offset = 0).

14. SHUTDOWN

- a) Click **Withdraw** in the Workflow Toolbar.
- b) Set X and Y offsets: 0, Scan angle: 0°
- c) Click **Navigate** and raise the scanner head so it is visibly clear of the surface. **LIFT the head up AT LEAST 1 cm above your sample.**

**CAUTION:**

Never withdraw samples without verifying that the tip has adequate clearance during the entire sample removal sequence.

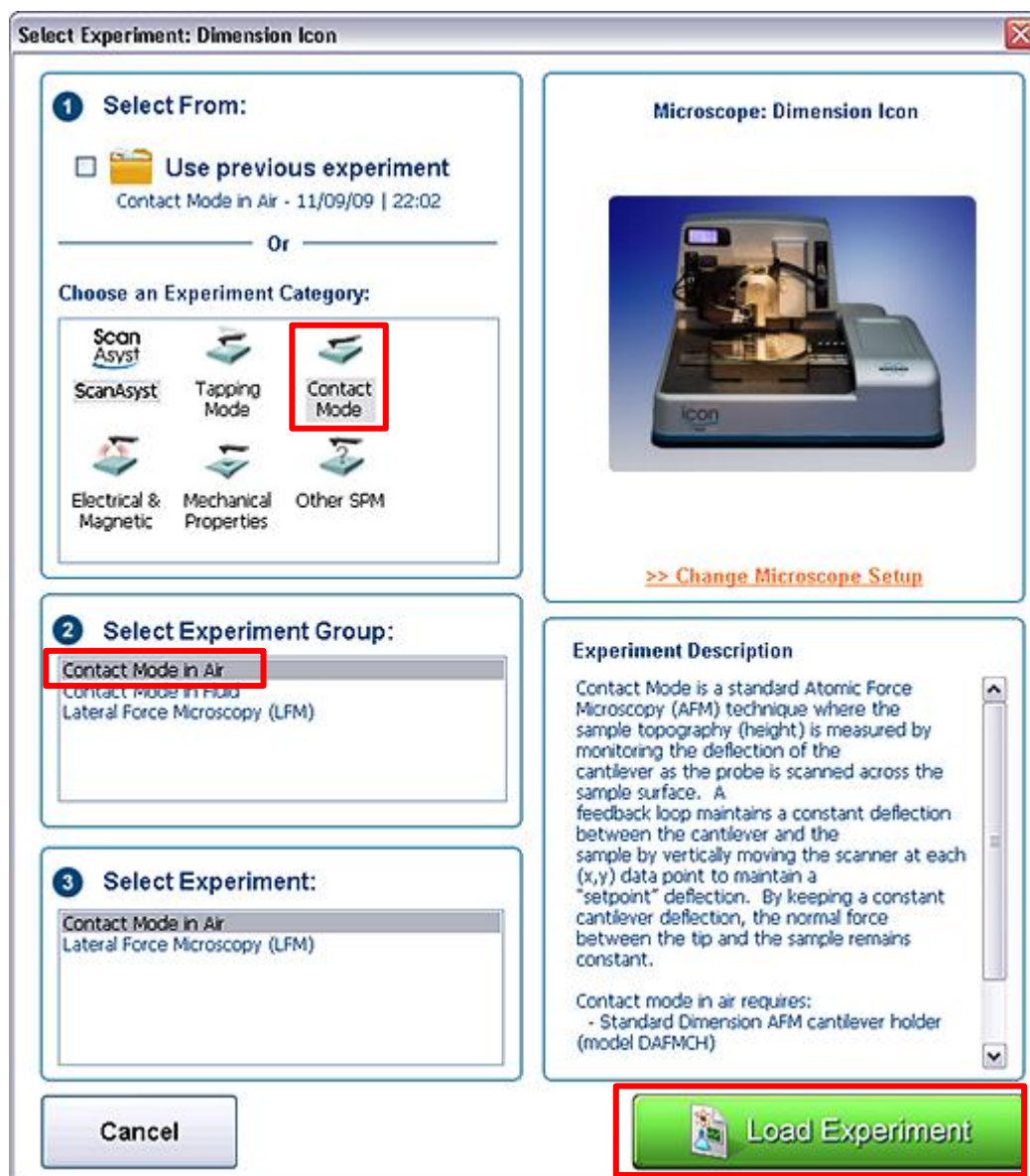
- d) With gloves on, open the window and rotate the stage.
- e) Turn off vacuum and collect sample.



- f) If necessary, clean the stage. Only use isopropanol on a KimWipe and never wet the stage directly with solvent. **DO NOT use acetone.**
- g) Close the glass window.
- h) Close the Nanoscope software.
- i) Turn off the N2 gas valve.
- j) Copy out your data from the Acer PC.
Use NanoScope Analysis Software to Modify image, e.g. Plane Fit, Flatten.
- k) Sign out of the logbook.

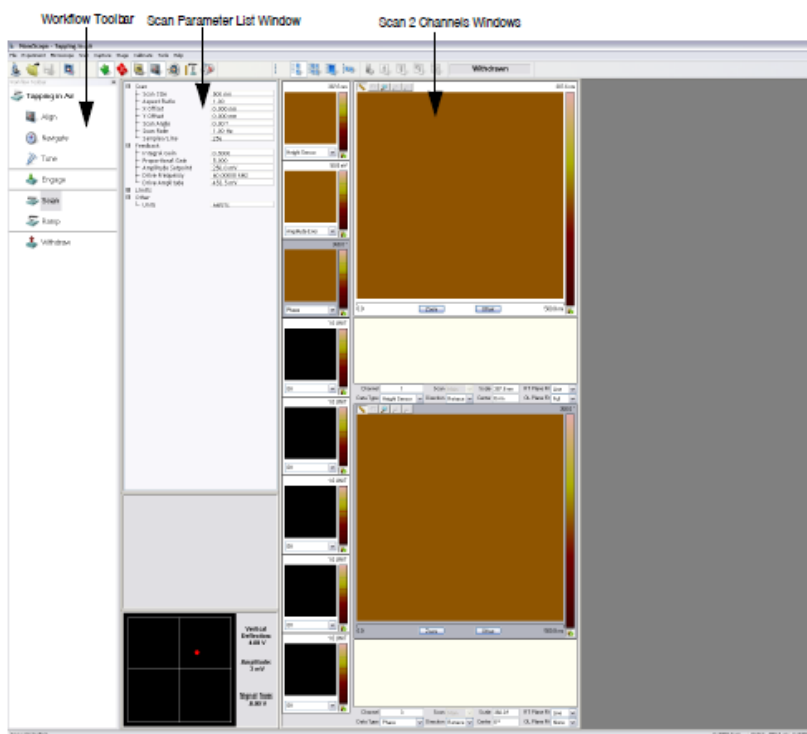
C) Contact Mode AFM

1. Put on gloves.
2. Turn on N2 gas valve.
3. Move and click the left mouse or turn on power button at bottom right corner of monitor.
4. Double click the **Nanoscope** software  on the desktop.
5. Select **Contact Mode** in the **Choose an Experiment Category** panel.

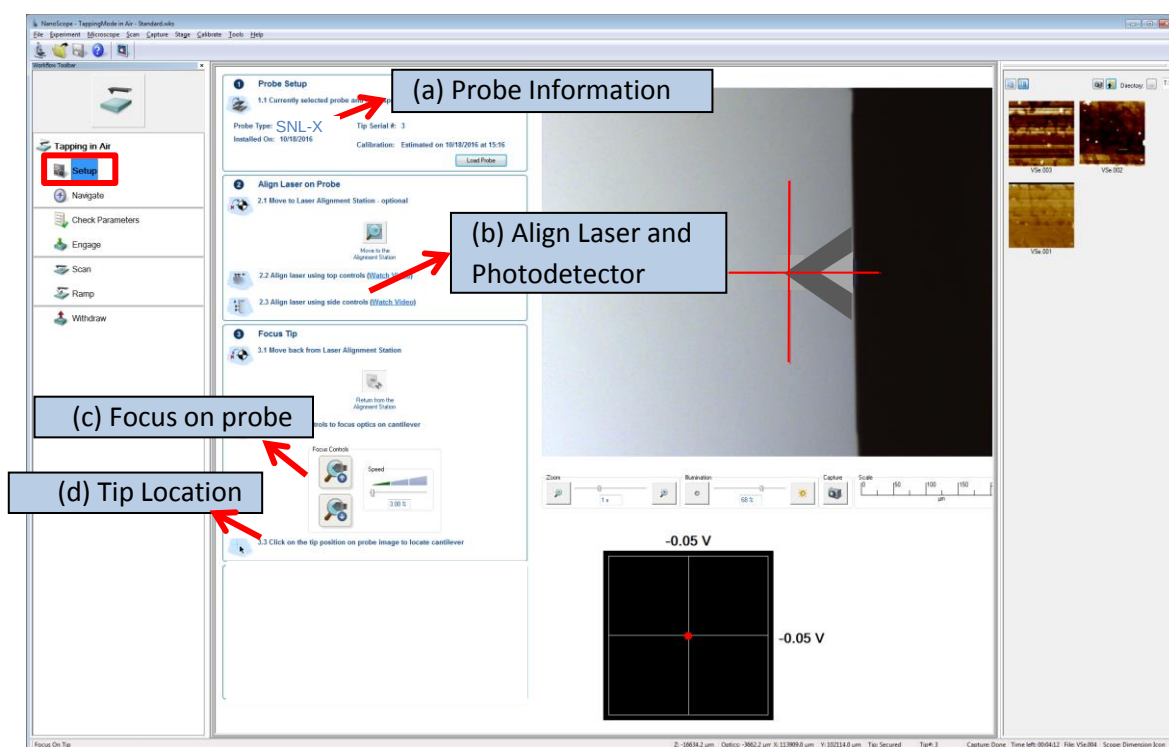


6. Select **Contact Mode in Air** in the **Select Experiment Group** panel and click **Load Experiment**.

The Workflow Toolbar, Scan Parameters List window and Scan 2 Channels windows will open.

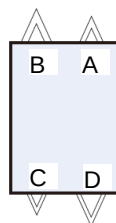


7. Click **Setup** in the Workflow Toolbar.
(Probe information>Align Laser & Adjust photodetector position>Focus Probe>Tip Location)



- (a) **Probe Information:** SNL-X (18 - 65 kHz, $k = 0.06$ to 0.35 N/m, tip radius < 10 nm)

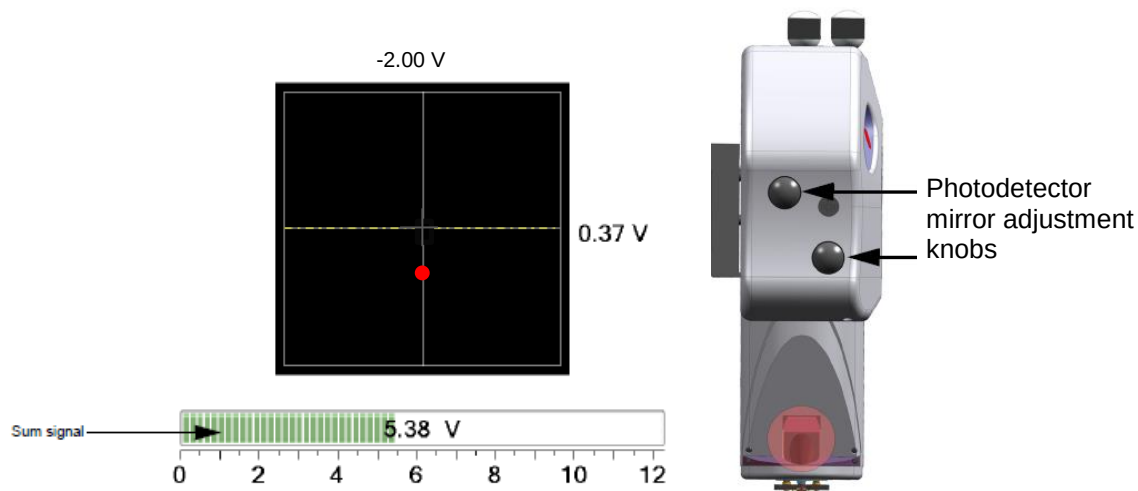
*Always ensure that you use the correct probe (Check with Lab Technologist)



Cantilever L & W	X	k (N/m)
L 120um, W 25um	A	0.35
L 205um, W 40um	B	0.12
L 120um, W 20um	C	0.24
L 205um, W 25um	D	0.06

(b) Align Laser and Photodetector (Do not turn the knobs too much as mechanism inside the head could break. **NEVER FORCE THE KNOBS.**)

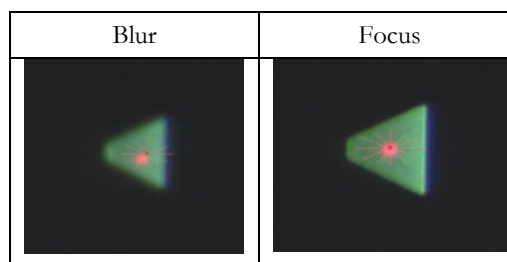
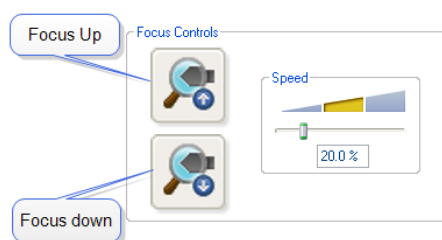
- i) *Laser Alignment.* Skip this step. It should have been pre-aligned. **Scanner head costs ~\$60K!**
Verify that the SUM signal is 4 - 6 V (laser power reaching the detector).
- ii) *Adjust photodetector.*
Verify/ use the adjustment knobs located on the left side of the SPM head, to set the Vertical Deflection to **-2 V**.



(c) Focus Probe (**must be done before focusing the optics on sample surface**)

[Optics is focused on probe tip so that Z position of tip is known. This is needed for successful tip engagement on sample surface.]

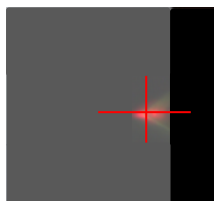
Adjust the speed accordingly and **Focus Up/down** till the cantilever end is sharp. Adjust illumination for better view.



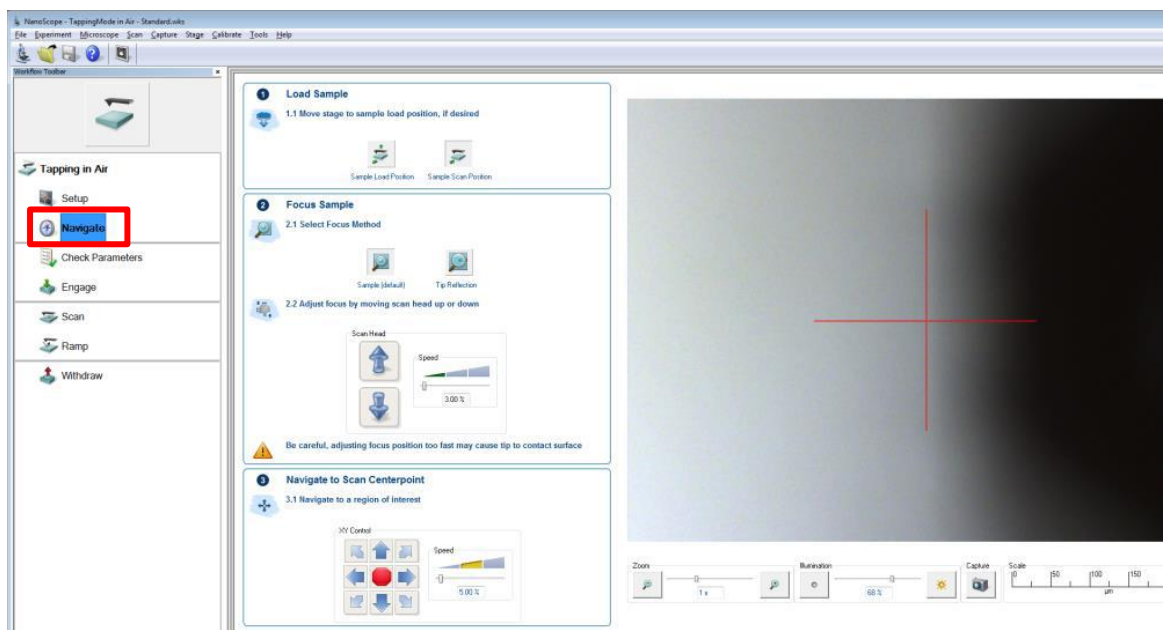
(d) Tip Location

Click to mark the crosshair on the position of the tip.

(When you leave the Locate tip mode, the optics move to a focus position, typically 1 mm below the tip.)



8. When the tip is in focus and centered in the field of view, click **Navigate** in the Workflow Toolbar. (Focus Sample → navigate to area of interest)

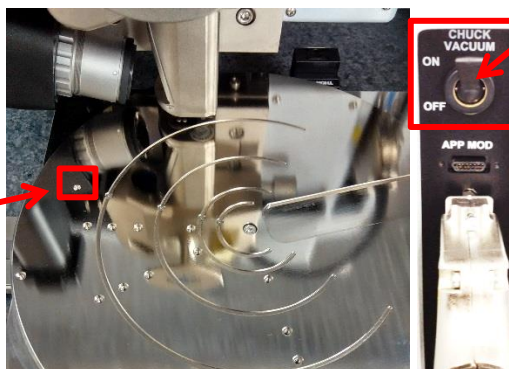


- a) Place sample on the stage and toggle 'ON' the chuck vacuum switch at the upper-right corner of the microscope, so that the sample is held down onto the stage.

*Ensure sample covers the vacuum hole fully and that the bottom of the sample is flat, so that sample is held down using vacuum.

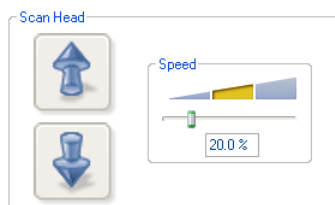
*If the sample size is big, use the other vacuum holes which are meant for big samples. (Ensure only 1 vacuum hole is open at all times; block all unused holes with screws.)

Vacuum holes
for small sample



Toggle switch to turn on vacuum
to hold down sample

- b) **Raise the SPM Head** using Scan Head "up arrow" to make sure that there is enough clearance space between the tip and the sample surface.



CAUTION:

When moving the SPM stage up and down, it is possible to crash the tip into the surface. To prevent a crash while focusing on the surface, watch the optical image and tip-to-sample proximity. The sample should be in focus when the tip is 1mm (1000μm) above the surface.

* For samples which are difficult to bring into focus, move to an edge of the sample, which is easy to find in the optical image, and bring the top of the edge into focus.

- (i) **Carefully** rotate the sample stage, like a turntable, so that the laser light falls on the sample surface.
- (ii) **Looking at tip-sample distance**, move the tip to **1-2 mm above the sample surface** by using the **Scan Head** arrows to focus on the sample surface or by rolling the trackball up or down while pressing the bottom-left button. [When you start seeing the red laser diffraction, on the optical image in the video panel, you are very close to finding the surface. **Decrease** the approach **speed** of the scanner head.]

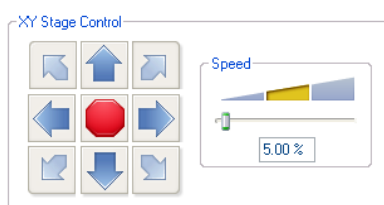


CAUTION:

Because the command **Focus Surface** moves the scanner vertically, be careful when making this adjustment to ensure that the tip does not hit the sample surface.

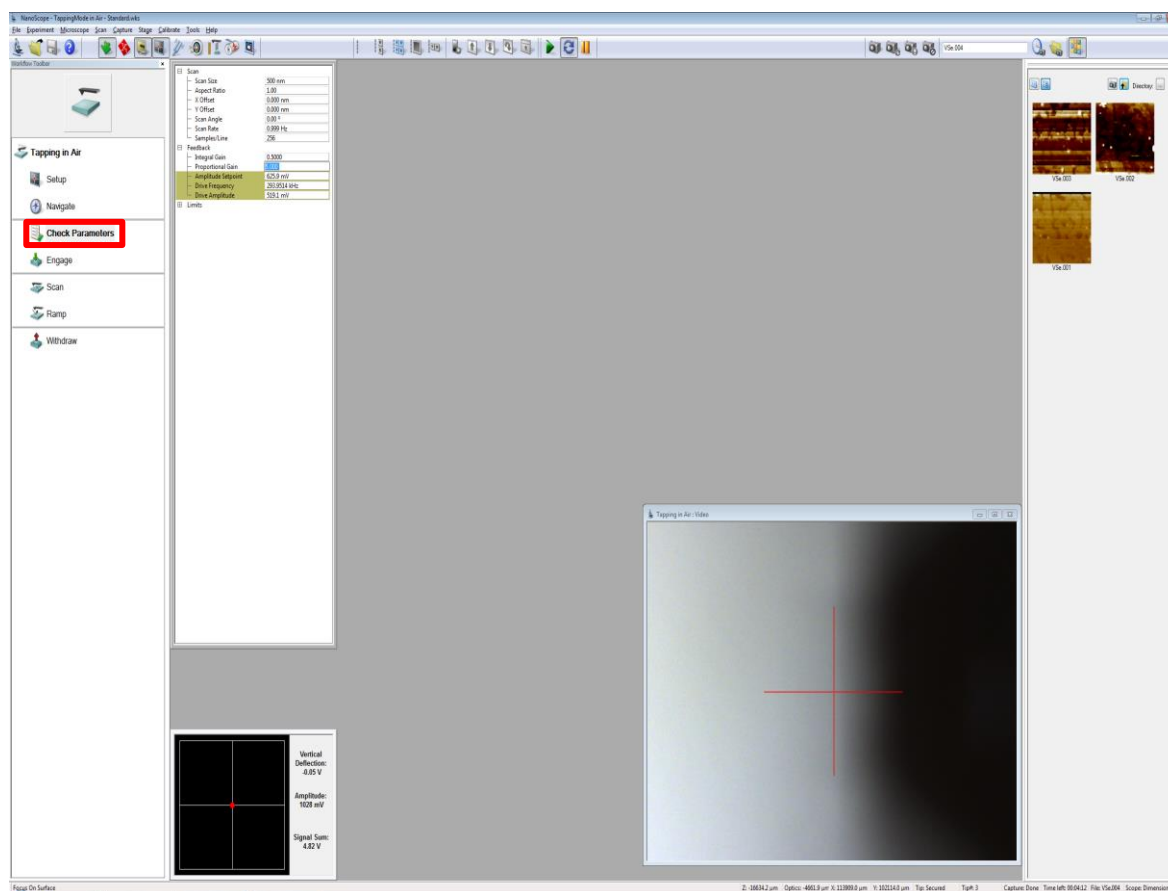
- c) Locate area of interest.

- (i) Move the X-Y stage to align the desired location on the sample under the crosshairs, either by using the XY Stage Control arrows or by rolling the trackball (without holding down any buttons).



- (ii) Verify that the surface is in focus.

9. Click **Check Parameters** in the Workflow Toolbar.



Scan	
Scan Size	500 nm
Aspect Ratio	1.00
X Offset	0.000 nm
Y Offset	0.000 nm
Scan Angle	0.00 °
Scan Rate	1.00 Hz
Samples/Line	256
Feedback	
Deflection Setpoint	0.4999 V
Integral Gain	2.000
Proportional Gain	5.000

Initial scan parameters for Contact Mode in Air

Bruker recommends always to engage with small scan sizes

- Scan Size: 500 nm
- X and Y Offset: 0
- Scan Angle: 0°
- Scan Rate: 1 Hz
- Integral gain: 2
- Proportional gain: 5

10. Click **Engage** in the Workflow Toolbar.

Scan lines appear in the Image Windows once the tip engages and scanning begins

- a) - Channel 1 Data Type: **Height Sensor**
(Data corresponds to the sensor measured change in piezo height needed to keep the cantilever deflection constant)
- Channel 2 Data Type: **Deflection Error**
(Data comes from the differential signal of the top and bottom photodiode segments relative to the Deflection setpoint. The error signal provides sensitive edge-detection and can be very helpful in visualizing fine details in topography that are difficult to see in regular height data)
- b) Click **AutoScale** icon  or set the Data Scale to a reasonable value.
- c) Adjust scan size to desired size.
- d) Optimizing Scan Parameters.

Check to see whether the trace and retrace lines in the **Scope** window are tracking each other well. They should have a similar shape, but they may not overlap each other horizontally or vertically. Adjust the **Deflection Setpoint**, **Integral Gain**, **Proportional Gain**, and **Scan Rate** parameters. Once the trace and retrace are tracking well, the tip is scanning the sample surface.

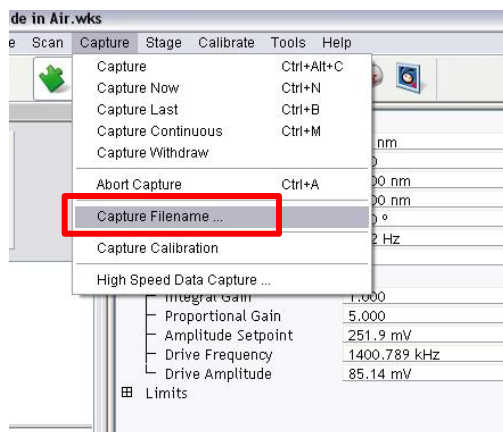
- (i) Adjust **Deflection Setpoint**.
The force applied by the tip on the sample increases as the setpoint is increased. To minimize the force on the sample:

Decrease the setpoint using the left arrow key on the keyboard, until the tip is no longer interacting with the surface (the trace and retrace lines are different and the Z Center Position move to the retracted limit).

Slowly increase the deflection setpoint using the right arrow key until features appear in the trace and retrace lines (the blue and red lines are very close or similar).
- (ii) Adjust **Integral** and **Proportional gains** to minimize error and noise. They must be high enough to force the feedback system to track the sample surface.

Increase **Integral Gain** until the piezo begins to oscillate, then eliminate the oscillations (high frequency wavy lines on image) by reducing the gain with 2 - 3 clicks of the left arrow. Repeat for **Proportional Gain** (usually set to 30%-100% more than **Integral Gain**)
- (iii) Adjust **Scan Rate**. Decrease the **Scan Rate** as the **Scan Size** is increased.
Use 1.5 – 2.5 Hz for large scans on samples with tall features. High scan rates help reduce drift, but only use them on flat samples with small scan sizes.

11. Image Capturing.

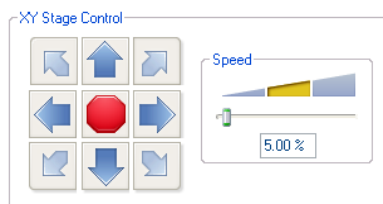


- Choose **Capture Filename** from the **Capture** menu and create your folder in T:\ drive.
*Make sure you have set the correct directory.
- Type your filename and click **Capture**
- Capture on** will be shown at the bottom right of the monitor.
*Capture will be completed when the full image is recorded under one set of parameters. If parameters are changed during one recording, the program will capture the next image.

12. To move to another area of interest, execute **Withdraw** in the Workflow Toolbar.

The SPM stops scanning and ascends.

- Use Scan Head up arrow (**Navigate** in Workflow Toolbar) if more clearance between the tip and sample is desired.
- Move the X-Y stage to align the desired location on the sample under the crosshairs, either by using the XY Stage Control arrows or by rolling the trackball (without holding down any buttons).



- Check surface is focused before clicking **Engage** (engage with small scan size, X and Y Offset = 0).

13. SHUTDOWN

- Click **Withdraw** in the Workflow Toolbar.
- Set X and Y offsets: 0, Scan angle: 0°
- Click **Navigate** and raise the scanner head so it is visibly clear of the surface. **LIFT the head up AT LEAST 1 cm above your sample.**

**CAUTION:**

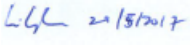
Never withdraw samples without verifying that the tip has adequate clearance during the entire sample removal sequence.

- With gloves on, open the window and rotate the stage.
- Turn off vacuum and collect sample.



- f) If necessary, clean the stage. Only use isopropanol on a KimWipe and never wet the stage directly with solvent. **DO NOT use acetone.**
- g) Close the glass window.
- h) Close the Nanoscope software.
- i) Turn off the N₂ gas valve.
- j) Copy out your data from the Acer PC.
Use NanoScope Analysis Software to Modify image, e.g. Plane Fit, Flatten.
- k) Sign out of the logbook.

Prepared By: Ng Soo Ngo

Approved By: A/P Tok Eng Soon  L.L. 21/5/2017